



CIMA Management Case Study

Mock Exam 4

Cartn

Suggested Solutions

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Task 1 - Solution

Sub-task (a) - 40%: Digital Data Sources

1ai - Identifying digital data sources to investment appraisals

A capital investment of H\$15 million represents a significant allocation of Cartn's resources and carries substantial risk given the innovative nature of digital coating. Standard investment appraisals often suffer from optimism bias or rely on static manufacturer estimates.

By leveraging digital data sources, the finance team can move toward a more dynamic, data-driven appraisal. This ensures that the NPV calculation reflects the actual operational environment of Cartn's 27 factories, rather than theoretical ideals.

Internal IoT and sensor data from prototype testing

Cartn could utilise real-time data captured by the Internet of Things (IoT) sensors currently installed on the Bio-Shield prototype machines.

Traditional estimates for operating costs are often based on best-case manufacturer specifications. By extracting data directly from the prototype's sensors, we can obtain precise measurements on electricity consumption per unit, the actual speed of the digital nozzles, and the exact downtime caused by the sensor calibration errors noted in the QC report.

Instead of asking a floor manager for a rough estimate of how often the machine stops, the finance team pulls the data directly from the prototype's cloud dashboard. This digital log records every micro-stop - even those lasting only 30 seconds - that a human operator might fail to record in a manual logbook.

For example, the IoT data could reveal that every time the machine switches from a 1-litre carton to a 0.2-litre carton, the digital nozzles require a 15-minute recalibration period due to sensor sensitivity. This results in an actual production speed of only 8,500 units per hour - 15% lower than the manufacturer's best-case claim.

This allows for a much more granular and accurate calculation of operating cash flows. It reduces the risk of the Board approving an investment based on underestimated maintenance or energy costs, ensuring the H\$15 million spend is anchored in operational reality.

External big data / Market sentiment analysis

Cartn could employ Web-Scraping or Social Media Analytics to quantify the Green Premium consumers are willing to pay. The financial viability of Bio-Shield depends on whether Cartn's customers can justify the higher cost of biodegradable packaging to retailers.

By analysing big data from global consumer trends and discussions, Cartn can model different demand elasticities.

Instead of relying on anecdotal evidence from the marketing director, the finance team can perform a sensitivity analysis on sales volumes based on verified market appetite. This ensures the revenue side of the NPV is not over-inflated by green-washing assumptions.

For example, Cartn could scrape retail pricing data from Harrrlandian supermarket websites to compare the price elasticity of eco-friendly milk brands versus standard brands.

If the digital data shows that green brands lose 40% of their volume when priced 10% higher than standard cartons, the finance team can identify a clear price ceiling for Bio-Shield. This moves the discussion from the Marketing Director's optimism to empirical evidence.

This data allows for a sensitivity analysis where Cartn can test the NPV's robustness against a smaller-than-expected green premium. This prevents Cartn from over-investing in capacity for a high-priced product that the average consumer - despite their best intentions - might find too expensive at the checkout.

External cloud-based commodity price indices

Cartn could look towards accessing real-time, cloud-based data feeds from global commodity markets for starch-based polymers vs. traditional polyethylene. By using digital price indices, Cartn can track historical volatility and forecast future price trends for these raw materials compared to fossil-fuel-based plastics.

For example, Cartn could integrate real-time API feeds from global agricultural commodity exchanges to track the price of industrial-grade corn starch - the primary raw material for the Bio-Shield coating.

By comparing this digital data to the historical price of crude oil (the driver for polyethylene), the finance team can identify the correlation risk. If the data shows that starch prices are highly sensitive to climate events or crop yields, Cartn can identify periods where their production costs might spike, even if oil prices remain low.

This allows the finance team to advise the Board on risk mitigation strategies, such as entering into forward contracts or fixed-price agreements with starch suppliers. By using

cloud-based data to prove that the H\$15 million investment is vulnerable to crop volatility, Cartn moves from a best guess to a sophisticated scenario plan that protects their long-term margins.

Sub-task (b) - 60%: Value management techniques

1bi - Recommending quality and value management techniques

The transition from traditional plastic lamination to digital bio-coating introduces both financial and technical complexities. To ensure the Bio-Shield project contributes to sustainable value creation, Cartn must balance cost efficiency with uncompromising quality.

Value Engineering will allow Cartn to optimise the product's design and cost base before full-scale manufacturing begins, while Total Quality Management (TQM) will foster a culture of right first time, addressing the technical failures identified in the recent QC report.

Application of Value Engineering (VE) to coating thickness

Cartn could apply value engineering to find the optimum coating thickness that maintains UHT integrity while minimising material use.

Value engineering focuses on providing the necessary function (protecting food at 135°C) at the lowest cost without sacrificing quality. Since the starch-based material is expensive and thins during UHT processing, Cartn engineers should collaborate with the finance team to redesign the application process - perhaps increasing the coating only at the critical seams while keeping it thinner on flat surfaces.

For example, Cartn should perform a functional cost analysis on the Bio-Shield carton to determine if a uniform coating is necessary across the entire surface.

Since the starch-based material is more expensive, the team should use the digital nozzles to apply a thicker, heat-resistant layer only to the seams and seals that failed at 135°C, while maintaining a thinner layer on the flat panels.

This prevents over-engineering and keeps the unit cost of the Bio-Shield carton competitive. By optimising the material distribution, Cartn can resolve the technical failures identified in the QC report without further bloating the production cost, ensuring the H\$15 million potential investment delivers a better outcome compared to a standard, non-optimised production line.

TQM: Quality at source and empowerment

Cartn could also implement quality at source by training factory operators to interpret the digital nozzle sensor data in real-time.

In this respect, total quality management (TQM) moves away from inspecting-in quality at the end of the line and builds it into the product. Operators should be empowered to halt production and recalibrate sensors the moment the digital dashboard shows a deviation. This shifts the culture from doing what we're told to one of professional ownership.

For example, Cartn could implement Statistical Process Control (SPC) dashboards at every Bio-Shield workstation, giving operators the autonomy to halt production the moment a sensor drift is detected.

By shifting the responsibility for quality from a separate inspection department to the operators themselves, Cartn addresses and potentially lowers the 4% leakage rate mentioned in the QC report before it becomes a mass-production failure.

This reduces internal failure costs of scrap and rework, while supporting Cartn's long-term sustainability strategy. By ensuring that quality is built-in rather than inspected-in, Cartn minimises the carbon footprint of wasted starch-materials and energy, helping the company reach its goal of reducing emissions below the current 24.0 g CO₂e per carton

Total Quality Management: Kaizen and continuous improvement

Another element of total quality management would be to establish a Kaizen Feedback Loop between the 800-strong consultancy division and the manufacturing factories. This draws on the total quality management philosophy of continuous improvement.

Cartn's consultants, who design packaging equipment at client sites, will see firsthand how the Bio-Shield cartons perform in real-world conditions. This data should be fed back to the director of product development to make incremental software updates to the digital nozzles.

For example, Cartn could establish a formal monthly Field-to-Factory forum where consultancy engineers share data on how Bio-Shield cartons perform during high-speed filling at client sites.

By Cartn using their 800 consultancy staff as eyes and ears in the field, they can identify real-world issues - such as the thinning at 135°C occurring more frequently on certain types of filling machinery - and feed this back to the manufacturing team for immediate, incremental adjustments to the digital nozzle settings.

This creates a competitive advantage that is difficult for the other five large manufacturers to replicate. By constantly refining the bonding of the bio-layers based on field data, Cartn ensures that its products are the most reliable in the market, strengthening its long-term partnership with major food producers.

Task 2 - Solution

Sub-task (a) - 60%: Business Models

2ai - Evaluate circular business model

The transition to a circular business model represents a shift from selling a physical product to managing a resource's lifecycle. For Cartn, this means moving away from the linear consumption of paperboard and aluminium and instead creating value by retaining ownership or oversight of the materials after the consumer has used them.

Resource efficiency and cost decoupling

The circular model creates value by decoupling Cartn's growth from the volatility of raw material prices. As in a linear model, Cartn has to take the market price for paperboard and aluminium. However, by recovering fibres through the 'Take-Back' scheme, Cartn creates a secondary internal supply chain.

For example, Cartn can use the Take-Back scheme to create a natural hedge against volatility in the global paper pulp market. By recovering fibres from used Bio-Shield cartons, Cartn builds a secondary inventory of raw materials. This allows the company to substitute expensive raw materials with recovered fibres in non-food-contact applications, such as the outer structural layers of frozen food tubs or transit packaging.

This decouples Cartn's profit margins from external commodity price shocks. If pulp and other raw material prices rise by 20%, Cartn's cost of sales only rises by a fraction of that amount because they are sourcing a portion of their raw material from their own recycled loop. This protects profit margins and ensures Cartn remains the lowest-cost provider even during resource shortages.

Brand differentiation and customer loyalty

This model enhances the perceived quality of Cartn's products, allowing for a price premium and deeper customer integration. By partnering with retailers for collection, Cartn becomes an essential part of the retailer's own Environmental, Social, and Governance (ESG) targets. This stickiness makes it harder for competitors to displace Cartn.

For example, Cartn can use the Bio-Shield range to transition from a transactional supplier to a strategic ESG partner for major retailers like Harr-Foods. By providing a Take-Back infrastructure and the associated environmental impact data, Cartn aligns its business model with the retailer's corporate sustainability goals.

This creates high switching costs for Cartn's customers, as once a retailer has integrated Cartn's recycling data into their public ESG reporting, they are less likely to switch to a lower-cost competitor who does not offer a circular solution.

This differentiation justifies a price premium for the Bio-Shield product, protecting Cartn's profit margins. Furthermore, the stickiness of the partnership ensures revenue stability through long-term exclusive contracts, which enhances the company's valuation and provides a sustainable competitive advantage over the other five large global manufacturers.

Mitigation of end-of-life sustainability risks

The circular business model addresses the recycling difficulty in Harrland and reduces the risk of future plastic taxes or legislative penalties.

Cartn's traditional cartons are a laminated sandwich of paper, aluminium, and plastic, and these are often dumped in a landfill site because local governments lack the facilities to separate them.

By taking responsibility for the end-of-life disposal, Cartn preemptively manages the risk that its products will be banned or heavily taxed by the Harrlanian government.

Legislators are increasingly shifting the financial burden of waste disposal from the taxpayer to the manufacturer. By taking responsibility for the end-of-life of the Bio-Shield cartons now, Cartn ensures its business model remains viable even if the government introduces heavy landfill levies or plastic taxes on non-recycled packaging.

This creates financial value for shareholders by shielding the company from unpredictable legal liabilities and tax penalties. It also addresses the recycling difficulty in Harrland, effectively future-proofing Cartn's market share against competitors who remain stuck in an unsustainable linear model that may eventually be banned or taxed out of existence.

Sub-task (b) - 40%: Managing stakeholders

2bi - Identify the key stakeholders and appropriate communication strategies

The transition to Bio-Shield and the circular model involves high levels of uncertainty. By using Mendelow's Matrix, Cartn can categorise stakeholders based on their power and interest. Effective communication is not a one-size-fits-all approach; it must be tailored to

address the specific fears and motivations of each group to maintain Cartn's social license to operate.

Environmental groups (High interest and High power)

These are key players who can damage Cartn's reputation if they perceive the scheme as greenwashing. The communication to these groups needs to be technical and transparent. They require proof that the downcycling is a transitional step toward full circularity.

For example, Cartn could engage environmental groups through a Technical Forum led by Wofai Eko, focusing on life cycle assessment transparency.

These stakeholders are highly influential and will quickly spot greenwashing. By showing that re-pulping for secondary packaging still reduces CO₂e by 30% compared to landfilling, Cartn provides these groups with hard data, turning them from critics into informed observers and reducing the risk of negative press.

This communication strategy turns a major reputational risk into a strategic alliance. If critical environmental groups publicly acknowledge that Cartn's scheme is a scientifically sound transition toward circularity, it neutralises press hostility and provides social proof to retailers and consumers that the Bio-Shield investment is a genuine sustainability breakthrough.

Small dairy farmers (High interest and Low power)

While these farmers are technically of low power on Mendelow's Matrix as they don't provide the massive volumes of the conglomerates, they still retain high interest because Cartn is likely their primary packaging partner.

Their primary fear is financial survival. Cartn could communicate with small dairy farmers by providing a tailored Bio-Shield Economic Impact Statement that demonstrates the long-term return on investment of switching to circular packaging.

These farmers fear that the H\$15m investment in Bio-Shield will be recovered through higher prices that favour large conglomerates. By using management accounting to show how the green premium paid by retailers will eventually flow back to the farmer, Cartn replaces fear with a clear business case for survival.

This strategy protects Cartn's customer diversity. By implementing a subsidised tiered subscription that will benefit the small dairy farmers, Cartn ensures that small farmers - who often have the strongest artisan brand appeal - stay within the Cartn ecosystem. This maintains the perceived quality of Cartn's customer base and prevents a public backlash that could damage the Group's reputation for fairness and sustainability.

Supermarket retailers (High power and High interest)

In Mendelow's Matrix, supermarket retailers are the ultimate key players with high power and high interest. Without them, the circular part of the business model fails because Cartn has no way to physically reach the consumer to "take back" the material

They are concerned with the operational footprint of collection points and the impact on their shoppers' experience. In terms of communication, Cartn could utilise a Key Account Management (KAM) strategy with major retailers, offering them a sustainability dashboard API as a digital incentive for participating in the Take-Back scheme.

Retailers are high-power stakeholders who view floor space as a premium asset. By providing a digital tool that allows them to gamify recycling for their shoppers and capture real-time data for their own sustainability reports, Cartn transforms the collection bin from an operational burden into a marketing and ESG asset for the retailer.

This ensures the logistical success of the circular model. By aligning Cartn's business model with the retailer's own strategic goals, it creates a barrier to entry for the other five major manufacturers who lack the digital integration to offer retailers a similar level of transparent, real-time sustainability data.

Task 3 - Solution

Sub-task (a) - 40%: Risk evaluation techniques

3ai - Recommend and explain the application of sensitivity analysis and scenario planning

The transition to a circular business model is no longer a theoretical exercise; the 22% pilot result and the H\$2.8 million IT overrun prove that the project is at a financial crossroads. To advise the Board, we must move away from best-case projections and use Sensitivity Analysis and Scenario Planning to determine the project's break-even survival points in a hostile regulatory environment.

Sensitivity analysis on the recovery rate

Cartn should use sensitivity analysis to determine the financial tipping point of the project now that this new information has come to light. Cartn currently has three conflicting numbers: the original 40% viability target, the 22% pilot reality, and the 50% tax exemption threshold. Sensitivity analysis will show how the NPV fluctuates as the project moves between these percentages.

Cartn should calculate the NPV at 22%, 40%, and 50% return rates. If the analysis shows that the project loses H\$100,000 in NPV for every 1% drop below the 50% tax-exemption threshold, the Board will realise that the 22% pilot result is not just a minor setback, but a deal-breaker that requires an immediate redesign of the collection logistics.

For example, the sensitivity analysis could reveal that at a 22% recovery rate, the project's NPV drops to -H\$3.5 million. This is primarily driven by the H\$500/tonne tax penalty incurred for failing to meet the 50% threshold, combined with the fact that Cartn's fixed logistics costs are being spread over a much smaller volume of recovered material. This turns the Bio-Shield from a high-margin innovation into a loss-making liability.

However, Cartn might find that if they spend H\$1 million on a massive marketing campaign and high-visibility Smart Bins to push the rate from 22% to 51%, the NPV jumps from a loss of H\$3.5m to a profit of H\$2m (a total value swing of H\$5.5m).

Scenario planning: Green leadership vs. The regulation trap

While sensitivity analysis tests the recovery rate in isolation, scenario planning combines multiple variables to create a narrative. Cartn should use scenario planning to model multiple futures where tax legislation, consumer adoption, and competitive response

intersect. This allows the Board to evaluate if the Bio-Shield range remains a viable project under different external pressures.

Cartn needs to see how their \$15 million investment performs if the government remains strict (50% threshold) while consumers remain indifferent and competitors slash prices on traditional packaging. This helps identify at what point the current strategy should be abandoned in favour of a plan B.

For example, Cartn should model a regulation trap scenario to test the resilience of the Bio-Shield business model against a combination of low consumer adoption and strict tax legislation. By combining variables - such as the 22% pilot return rate and the H\$500 per tonne plastic tax - the Finance team can visualise a world where Cartn is double-penalised: once by the tax and once by the high fixed costs of the logistics infrastructure.

This stress test is more valuable than a simple sensitivity analysis because it accounts for the interdependency of external risks. This enables strategic agility. If the regulation trap scenario shows that the project becomes cash-flow negative within two years, the Board can proactively decide to pivot the project towards an alternative viable model.

For example, if the 50% target is unachievable, Cartn should stop the H\$2.8m IT spend and instead re-invest in R&D to make Bio-Shield 100% compostable, seeking a different tax exemption that doesn't rely on consumer behaviour. This protects shareholder value by preventing sunk cost fallacy and ensuring that Cartn's capital is allocated to the most resilient long-term solution.

Sub-task (b) - 60%: Project management tools

3bi - Evaluate and recommend three project management tools or techniques

The Bio-Shield IT project is currently suffering from a governance void. With the budget ballooning from H\$1.2m to H\$2.8m and unauthorised email-based scope changes, the project is at risk of total failure. Cartn must implement rigid controls to realign the Marketing, Logistics, and IT departments.

A Formal Change Control Board (CCB)

Cartn must immediately implement a Change Control Board (CCB) to halt scope creep from the Marketing department. The project committee revealed that Haitham Amer is requesting Individual Carbon Credits without a formal cost assessment.

Every project is managed within the triple constraint of time, cost, and quality. If you increase the scope by adding personalised carbon tracking, it must impact either the cost or the time.

By having a change control board in place - consisting of Mario (Finance), Wofai (Product), and the Project Lead - they must approve or reject every change based on its impact on the triple constraint of time, cost, and quality.

For example, the current H\$1.6 million budget overrun suggests that project boundaries are being ignored. By requiring a formal impact assessment for every new feature - such as the individual carbon credits - the CCB ensures that the Board only approves changes that have a clear return on investment or are essential for hitting the 50% regulatory recovery target.

This restores financial discipline to the project. It prevents death by a thousand cuts, where small, unauthorised changes aggregate into massive delays. The CCB ensures the IT system is ready for the Bio-Shield launch, maintaining the project's viability and preventing further erosion of the H\$15 million capital investment.

Critical Path Analysis (CPA) & Resource levelling

The critical path is the longest sequence of dependent tasks in a project. If any task on this path is delayed by a single day, the entire project is delayed by a day.

In Cartn's case, the IT/Logistics deadlock is likely sitting right on the critical path. Tasks off the critical path have float (i.e. extra time), meaning delays there don't hurt the final deadline. However, the IT Tracking System is a prerequisite for the Bio-Shield launch; therefore, its delay is currently pushing back the H\$15 million revenue stream.

In this situation, Cartn should use critical path analysis (CPA) to pinpoint the specific technical dependencies causing the 3-month delay and apply Resource Levelling to recover the schedule.

In a complex project like the Bio-Shield infrastructure, many tasks can happen in parallel. By identifying the critical path, the project manager can see which tasks have float and which are bottlenecks. This allows the Board to move resources - such as personnel or equipment - away from non-urgent tasks and toward the critical IT/Logistics interface.

This enables schedule recovery without necessarily increasing the H\$2.8 million budget. By moving internal experts to solve the deadlock in real-time, Cartn reduces the time-to-market for Bio-Shield. This ensures the company can begin capturing the Green Premium and meeting the 50% tax-exemption target sooner, protecting the project's overall profitability.

A Risk Register & Red-Amber-Green (RAG) Reporting

A risk register quantifies threats using impact and probability. Since the 22% pilot result makes the plastic tax a high-impact, high-probability event, it should be classified as a critical risk.

The RAG (Red-Amber-Green) status provides an instant visual hierarchy:

- **RED:** Immediate threat to the project's viability (i.e. the 50% Tax Threshold). This would require Board-level intervention.
- **AMBER:** Potential for delay or cost overrun (i.e. the IT/Logistics deadlock). Needs departmental management to resolve.
- **GREEN:** Managed risk within acceptable tolerances (i.e. sourcing paperboard).

The risks identified in the register have an owner assigned to each one, therefore making key individuals accountable. By implementing a risk register with RAG reporting, Guiquan Li doesn't have to hunt through 50-page-plus reports; he can look at the dashboard and see exactly where the H\$15 million capital is at risk and who is responsible for resolving the risk.

In this case, it's highly recommended that Cartn implement a dynamic risk register utilising Red-Amber-Green (RAG) reporting to categorise and manage the high-impact regulatory and operational threats facing the Bio-Shield project.

A project of this scale often suffers from information overload, so RAG reporting distils complex issues, like the 50% tax-exemption threshold, into a simple visual format that forces the Board to focus on the most critical threats first. Assigning a specific risk owner (i.e. Alison for logistics), it creates clear accountability for mitigation rather than allowing risks to remain unaddressed in project meetings.

By regularly reviewing the RED risks, Guiquan Li can ensure that capital is being deployed to the most urgent problems, such as fixing the 22% return rate. This prevents a slow, invisible failure of the project and ensures that Cartn's H\$15 million investment is actively defended against external regulatory shocks.

Task 4 - Solution

Sub-task (a) - 60%: Accounting Treatment

4ai - Accounting treatments for extruders and site clean-up (IAS 36, IAS 16, IAS 37)

The transition of Plant Alpha is a significant accounting event that requires Cartn to move beyond routine bookkeeping and apply prudence and fair presentation under IFRS. As Cartn decommissions legacy plastic assets and publicly commits to environmental restoration, its financial statements must accurately reflect the economic reality of these 'sunsetting' operations.

Failing to recognise these impairments and provisions now would mislead our shareholders regarding the true cost of our transition to Bio-Shield.

Traditional extruders: Accounting treatment

Cartn must immediately conduct an Impairment Test under IAS 36 as the Board's decision to cease plastic production and move to Bio-Shield technology is a clear indicator of impairment. Cartn must compare the H\$8.4 million carrying amount to the asset's recoverable amount, i.e. the higher of its fair value less costs to sell and its current value in use.

Given the impending obsolescence and the scrap quote of only H\$200,000 in 18 months, its current fair value on the open market will be very low. Therefore, Cartn must calculate the value in use based on the cash flows generated by the machines over their remaining 18 months. If that value in use is, for instance, H\$5 million, Cartn must recognise an immediate impairment loss of H\$3.4 million in the statement of profit or loss.

Following that, Cartn must perform a revision of useful life and residual value under IAS 16. In Cartn's case, the shift from plastic to Bio-Shield is a significant change in the expected pattern of consumption, which triggers a mandatory revision of its usefulness.

The remaining H\$5 million carrying amount must be depreciated down to its H\$200,000 residual (scrap) value over the new 18-month timeline. This depreciable amount of H\$4.8 million equates to a H\$266.6k monthly charge to the profit and loss.

All in all, Cartn's impact on their P&L will be a significant one-off impairment charge of H\$3.4 million and higher monthly depreciation expenses of H\$266.6k that will sharply reduce operating profit in the short term. Meanwhile, the carrying amount of non-current assets will decrease rapidly, ensuring that Cartn's balance sheet isn't inflated by obsolete technology as they pivot to the Bio-Shield technology.

Under IAS 36 and IAS 16, Cartn are required to revise the useful life now, otherwise it would face adverse audit opinions and potential fines by the Harrlandian Stock Exchange and financial authorities. Cartn would also be hit with a massive loss on disposal in 18 months when they throw the machines away. Revising the life now effectively smooths that hit over the next 18 months and keeps their professional reputation intact.

Site decontamination: Recognition of a provision

Cartn must recognise a provision for the H\$3.2 million clean-up cost under IAS 37 Provisions, Contingent Liabilities and Contingent Assets.

While there is no legal requirement to clean the site yet, the CEO's televised commitment has created a Constructive Obligation. Cartn has signalled to the public that it will accept this responsibility, leaving the company with no realistic alternative but to settle the obligation.

With that in mind, Cartn must record the obligation at its present value. Since the H\$3.2 million is payable in two years, we would discount this figure using a pre-tax discount rate. If we assume a discount rate of 5%, Cartn would record an initial provision of approximately H\$2.9 million today.

In terms of impact on the financial statements, Cartn will recognise a H\$2.9 million liability immediately that will increase Cartn's debt-to-equity ratio. If Cartn has strict bank covenants regarding debt levels, this could technically put those covenants at risk.

On the profit and loss statement, Cartn will recognise an immediate H\$2.9 million expense for the initial provision. Additionally, as the payment date moves closer, Cartn will recognise a finance cost (unwinding of the discount), which will marginally impact profit over the next two years.

By recognising this now, Cartn are essentially pre-paying the reputational cost of the transition. It shows investors that their green commitment isn't just a marketing slogan from Haitham Amer, but a recorded financial reality that Cartn are prepared to fund.

The combined impact of these treatments will be a short-term reduction in reported profits and a decrease in net assets. However, this is essential for the fair presentation of the accounts. By clearing the decks of legacy plastic-related costs now, Cartn ensure that when the Bio-Shield range launches in full, its future profitability is not masked by the

lingering costs of obsolete machinery or undisclosed environmental liabilities. This builds trust with investors who are specifically looking at their ESG (Environmental, Social, and Governance) performance.

Sub-task (b) - 40%: Stakeholder management

4bi - Recommending resolution strategies for stakeholder conflict

The current impasse between the sales and production departments threatens the operational success of the Bio-Shield rollout. While Haitham Amer is understandably protective of his market share, and Alison Cruikshank is rightly focused on legal compliance, their inability to collaborate has created a strategic bottleneck.

Confronting / Problem-solving (The win-win approach)

To resolve this, the project sponsor and lead must move from a compromise mindset to an integrating strategy, aligning both leaders behind the singular goal of long-term success of the Bio-Shield project.

Sales need a marketable price, and production needs regulatory compliance. These are not mutually exclusive. Instead of a compromise, where both lose, the Board should force a deep-dive into the Bio-Shield cost structure to find value engineering opportunities.

For example, Cartn could task the consultancy sales team to work with production to see if the advisory service can be bundled with the high-cost Bio-Shield cartons. Instead of a compromise that might lead to subpar product quality, integration seeks to satisfy the underlying interests of both departments. By leveraging Cartn's 800 engineers to provide free transition support to customers, they provide a value-added service that justifies the price premium to the customer.

This gives the sales team a win, i.e. a sellable package and also production a win as compliance is met. By reframing the 25% price increase as an investment in regulatory insurance, sales can use production's compliance requirements as a competitive advantage rather than a burden. This turns a departmental conflict into a unified market strategy.

Alignment of incentives (Creating goal congruence)

The Board could implement a Shared Reward System and Joint KPIs that tie the bonuses of both Haitham (Sales) and Alison (Production) to the successful and profitable transition of the Bio-Shield range.

Currently, the conflict is fueled by a lack of goal congruence. Haitham is likely measured on sales volume and revenue (which is harder with a 25% price hike), while Alison is measured on regulatory compliance and production efficiency. They are literally being paid to fight each other. By creating joint KPIs, the Board forces them to collaborate because one cannot achieve their bonus without the help of the other.

For example, instead of separate departmental targets, the Board could introduce a Bio-Shield Profitability & Compliance Tier for the annual bonus targets:

1. The Sales Metric: Haitham's bonus is no longer based on total volume, but on the contribution margin of Bio-Shield units. This encourages him to stop complaining about the price and start finding high-value customers who appreciate the plastic-free brand.
2. The Production Metric: Alison's bonus is tied to target costing. She is rewarded if she can work with her engineers to reduce the 25% cost premium through better manufacturing efficiencies over the next 18 months.
3. The "Clawback" Clause: If the company fails to meet the Harrland government plastic ban deadline, then both directors lose 50% of their total bonus.

Once Haitham and Alison realise their personal financial outcomes are linked, the blame game stops. Haitham will stop viewing Alison as a saboteur and start viewing her as a partner who can help him lower costs to make the product more sellable. Conversely, Alison will view Haitham's market feedback as essential data for her production planning. This turns the current dysfunctional conflict into functional collaboration focused on the company's bottom line.

Marking Matrix

Task	Topic	Core activity	%	Marks	Total
1a	Identifying digital data sources to investment appraisals	A	40	10	25
1b	Recommending quality and value management techniques	C	60	15	
2a	Evaluate circular business model	A	60	15	25
2b	Identify the key stakeholders and appropriate communication strategies	E	40	10	
3a	Recommend and explain the application of the sensitivity analysis and scenario planning.	D	40	10	25
3b	Evaluate and recommend three project management tools or techniques.	B	60	15	
4a	Applying IAS 36, IAS 16 and IAS 37 accounting standards	D	60	15	25
4b	Recommending resolution strategies for stakeholder conflicts	E	40	10	
					100

Marking guidance

Task 1		
(a)	A. Evaluate opportunities to add value	40%
(b)	C. Manage performance and costs to create sustainable value	60%

A. Evaluate opportunities to add value		40%
1ai - Identifying digital data sources		10 marks
	No rewardable material	0
Level 1	One digital data source provided, no or little application to the scenario.	1-3
Level 2	One or two digital data sources provided, with some application to the scenario.	4-6
Level 3	Three or more digital data sources provided, with strong application to the scenario.	7-10

C. Manage performance and costs to create sustainable value		60%
1bi - Recommending quality and value management techniques		15 marks
	No rewardable material	0
Level 1	Weak recommendation of up to one technique. No application to scenario.	1-5
Level 2	One or two recommendations provided, with some application to the scenario.	6-10
Level 3	Two or more recommendations provided for both Value Engineering and TQM, with strong application to the scenario.	11-15

Task 2		
(a)	A. Evaluate opportunities to add value	60%
(b)	E. Manage internal and external stakeholders	40%

A. Evaluate opportunities to add value		60%
2ai - Evaluate circular business model		15 marks
	No rewardable material	0
Level 1	Weak evaluation of business model, with little to no application to scenario.	1-5
Level 2	Good evaluation of the business model and explanation of sustainable value, with some application to the scenario.	6-10
Level 3	Strong, comprehensive evaluation of the circular business model and clear explanation of sustainable value creation, with strong application to the scenario.	11-15

E. Manage internal and external stakeholders		40%
2bi - Identify the key stakeholders and appropriate communication strategies		10 marks
	No rewardable material	0
Level 1	Surface-level suggestion of the key stakeholders.	1-3
Level 2	Stakeholders identified with some reasoning and application to the scenario.	4-6
Level 3	Perceptive identification of the stakeholders with strong application to the scenario.	7-10

Task 3		
(a)	D. Measure performance	40%
(b)	B. Implement senior management decisions	60%

D. Measure performance		40%
3ai Recommend and explain the application of sensitivity analysis and scenario planning		10 marks
	No rewardable material	0
Level 1	Weak recommendation(s) with no application to the scenario	1-3
Level 2	Clear recommendation(s) with some application to the scenario	4-6
Level 3	Strong recommendation(s) with strong application to the scenario	7-10

B. Implement senior management decisions		60%
3bi - Evaluate and recommend three project management tools or techniques.		15 marks
	No rewardable material	0
Level 1	Evaluates and recommends one project management tool or technique. The explanation is weak or generic, with little to no application to the scenario.	1-5
Level 2	Evaluates and recommends two project management tools or techniques. The explanation is clear, with some application to the scenario.	6-10
Level 3	Evaluates and recommends three appropriate project management tools or techniques. The explanation is comprehensive, with strong, in-depth application to managing the project's specific risks and challenges.	11-15

Task 4		
(a)	D. Measure performance	60%
(b)	E. Manage internal and external stakeholders	40%

D. Measure performance		60%
4ai - Applying IAS 36, IAS 16 and IAS 37 accounting treatments		15 marks
	No rewardable material	0
Level 1	Weak application of accounting standards. No recommendation or application to scenario.	1-5
Level 2	Good application of accounting standards with some recommendation or application to scenario.	6-10
Level 3	Strong application of accounting standards with good recommendation or application to scenario.	11-15

E. Manage internal and external stakeholders		40%
4bi - Recommending resolution strategies for stakeholder conflict		10 marks
	No rewardable material	0
Level 1	Weak evaluation of stakeholder conflict. No recommendation or application to scenario.	1-3
Level 2	One or two strategies evaluated with a recommendation and some application to the scenario.	4-6
Level 3	Two clear conflict resolution strategies evaluated and recommended with strong application to the scenario.	7-10